CS6290 — Individual Evidence Pack (Milestone 3)

Student Information

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- **Group Number / Project Title:** Group 2 / Polymarket Signal Analysis — Empirical Detection of Unusual Odds Movements
- **Milestone:** Milestone 3
- **Date:** 29/03/2026

1) What I Contributed

Branch 1: Anomaly Detection Visualization Integration

- **Three-Layer Drill-Down Interface**: Implemented a hierarchical visualization system for market stress anomaly detection with three progressive levels of detail, following the "Top-Down" design approach.
- **Market Overview Layer**: Developed a top-level dashboard displaying cross-market stress event rankings and health comparisons using bar charts and data tables, sourcing data from market_stress_summary and all_stress_events files.
- **Single Market Events Layer**: Created an intermediate view featuring event attribution pie charts and interactive bubble charts correlating stress scores with trading volumes, using stress_events_enriched data.
- **Event Diagnostics Layer**: Built a detailed diagnostic panel with four synchronized subplots showing stress scores, trading activity, liquidity status, and price trends within event time windows, with 10-minute extended time range for context.

Branch 2: Sniper Detection Frontend Showcase

- **Card-Based UI Implementation**: Designed and implemented a "Detective Report" card-based interface for displaying sniper attack detection results with risk-level color coding (red for high risk, orange for medium, green for low).
- **Verified Cases Display**: Created an expandable card component for showcasing three verified sniper attack cases (rank 2, 3, 4) with complete transaction details and attack window visualizations.
- **Candidate Ranking System**: Implemented a paginated list view (5 candidates per page) for browsing sniper candidates sorted by suspicious rank, featuring anomaly score distribution histograms.
- **Data Integration**: Integrated sniper detection datasets including detailed_cases.json, strict_sniper_candidates.csv, all_cases.json, and attack window plot images.

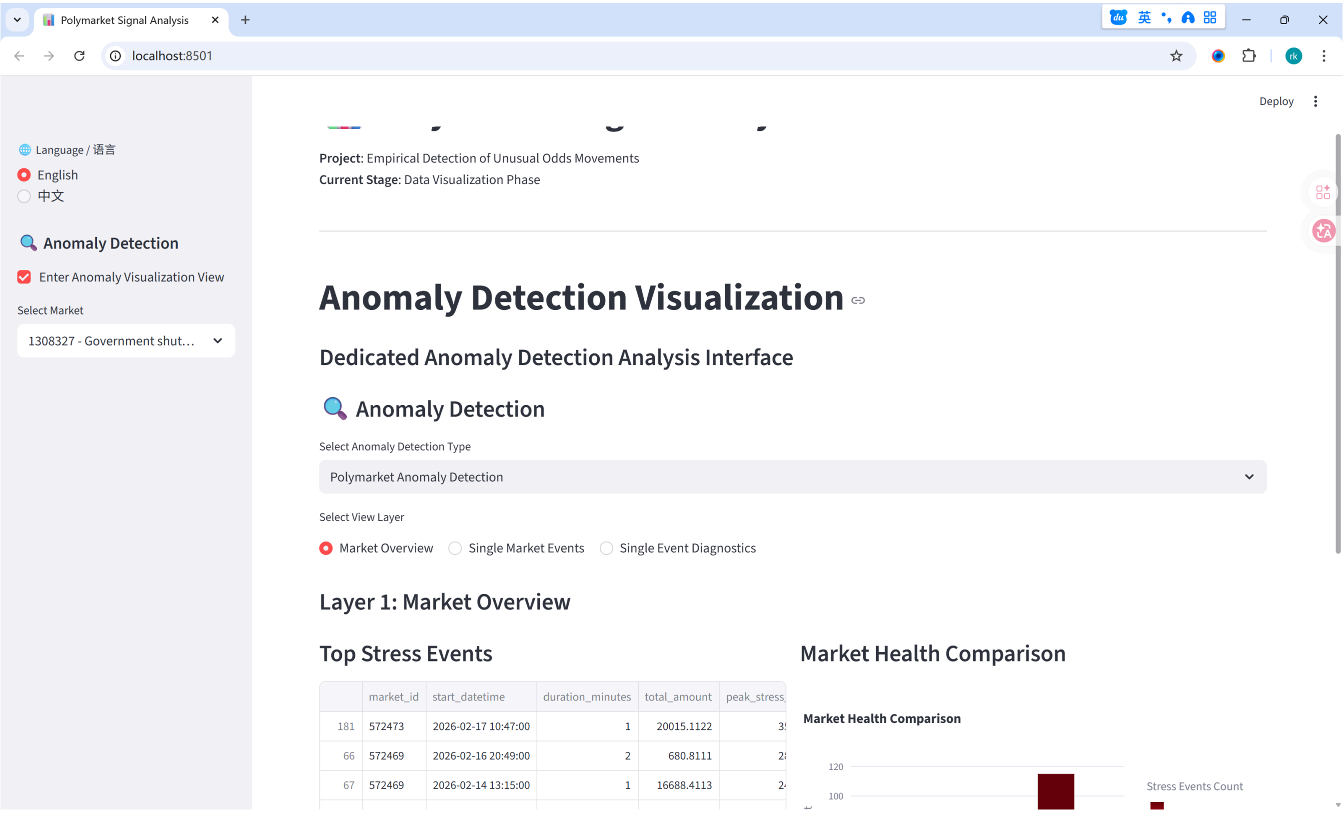
Branch 3: Data Loading Infrastructure

- **Extended Data Loader Functions**: Added specialized loading functions for anomaly detection outputs including bucket features, bucket scores, market stress summaries, and stress event collections.
- **Multi-Source Data Support**: Implemented unified data access patterns for both market stress anomaly results and sniper detection outputs stored in different formats.
- **Caching Optimization**: Applied Streamlit caching decorators to all new data loading functions to maintain dashboard performance with large anomaly detection datasets.

2) Evidence

#	Evidence type	Link / Reference	What this shows
1	Repository	https://github.com/zrk494/ProjectOfPrivacy-enhancingTech/tree/main/submodules/visualize	The central repository containing the visualization module with integrated anomaly detection interface.
2	Screenshot	The image below	Implemented three-layer drill-down anomaly detection visualization with synchronized multi-metric charts.

- **Anomaly Detection Visualization**



3) Validation Performed

- **Multi-Layer Navigation Testing**: Verified seamless transitions between the three visualization layers (market overview, single market events, event diagnostics) with consistent data state preservation.

- **Cross-Market Data Loading**: Confirmed successful loading and display of anomaly detection results across multiple market datasets with proper datetime parsing and formatting.
- **Sniper Detection Data Integration**: Validated correct parsing and display of JSON-based sniper detection results including transaction details and attack window images.
- **Visualization Component Validation**: Tested all Plotly chart components including pie charts for attribution, bubble charts for event correlation, and synchronized subplots for time-series diagnostics.
- **Pagination Functionality**: Verified the paginated display of sniper candidates with proper session state management for page navigation.
- **Bilingual Interface Testing**: Confirmed all new interface elements and labels display correctly in both English and Chinese language modes.

4) AI Usage Transparency

- **AI tool(s) used:** Used **DeepSeek** for researching visualization design patterns for anomaly detection dashboards. Used **Gemini** to assist in structuring the three-layer drill-down interface and refining documentation.
- **One AI output I rejected (and why it was wrong, risky, or insufficient):**
  - **Rejection**: Rejected the suggestion to use a single complex dashboard with all metrics visible simultaneously.
  - **Correction**: Adopted a progressive disclosure approach with three distinct layers to prevent information overload and improve user navigation flow.

5) Risk

- **Data synchronization between anomaly detection pipeline and visualization.**
  - *Detail*: Changes to output file formats in the algorithm submodule may require corresponding updates to data loader functions; established clear path conventions to mitigate this risk.